

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406389
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Ghannam; Mahmoud Yousef et al.

Vehicle space morphing

Abstract

A method for determining spatial availability for a compartment of a vehicle includes generating, via at least one time-of-flight sensor, a first point cloud representing the compartment of the vehicle, the first point cloud including three-dimensional positional information of the compartment. The method further includes calculating, via processing circuitry in communication with the at least one time-of-flight sensor, spatial information corresponding to a target configuration for the compartment. The method further includes comparing, via the processing circuitry, the spatial information to the first point cloud. The method further includes determining an availability of the target configuration based on the comparison of the spatial information to the first point cloud.

Inventors: Ghannam; Mahmoud Yousef (Canton, MI), Van Wiemeersch; John Robert (Novi, MI)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

Family ID: 92543851

Assignee: Ford Global Technologies, LLC (Dearborn, MI)

Appl. No.: 18/122292

Filed: March 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240312043 A1	Sep. 19, 2024

Publication Classification

Int. Cl.: G06T7/70 (20170101); G01S17/89 (20200101); G06V20/59 (20220101)

U.S. Cl.:

CPC **G06T7/70** (20170101); **G01S17/89** (20130101); **G06V20/59** (20220101);
G06T2207/10028 (20130101); G06T2207/30268 (20130101)

Field of Classification Search

CPC: G06T (7/70); G06T (2207/10028); G06T (2207/30268); G06T (7/62); G06T (19/003);
G06T (7/73); G06T (2210/44); G01S (17/89); G06V (20/59)

USPC: 382/100

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10836401	12/2019	Christensen	N/A	B60W 50/0098
11062582	12/2020	Van Wiemeersch	N/A	G08B 29/24
11067983	12/2020	Kentley-Klay et al.	N/A	N/A
2018/0035248	12/2017	Soave	N/A	H04W 4/025
2019/0339087	12/2018	Jindal	N/A	G06N 20/00
2020/0378775	12/2019	Beaurepaire	N/A	G01C 21/3423
2020/0389760	12/2019	Ghannam et al.	N/A	N/A
2020/0406902	12/2019	McBride	N/A	N/A
2022/0237965	12/2021	Vaccariello	N/A	H01Q 1/3241
2024/0346938	12/2023	Rozenberg	N/A	G08G 5/32

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110895724	12/2019	CN	G06Q 10/04

Primary Examiner: Grant, II; Jerome

Attorney, Agent or Firm: Price Heneveld LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to space morphing for vehicles and, more particularly, relates to a detection system for determining spatial availability of a vehicle compartment.

BACKGROUND OF THE DISCLOSURE

(2) Conventional techniques for modifying vehicle spaces may involve complex analysis of the space. Methods of determining the availability for these changes are typically challenging to carry out due to the lack of accurate three-dimensional information about the vehicle spaces and limited visualization or analysis of a transformation. There is a need for a more dynamic process for optimizing vehicle spaces.

SUMMARY OF THE DISCLOSURE

(3) According to a first aspect of the present disclosure, a method for determining spatial

availability for a compartment of a vehicle includes generating, via at least one time-of-flight sensor, a first point cloud representing the compartment of the vehicle. The first point cloud includes three-dimensional positional information of the compartment. The method further includes calculating, via processing circuitry in communication with the at least one time-of-flight sensor, spatial information corresponding to a target configuration for the compartment. The method further includes comparing, via the processing circuitry, the spatial information to the first point cloud. The method further includes determining an availability of the target configuration based on the comparison of the spatial information to the first point cloud.

(4) Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features: determining the target configuration based on an occupancy condition of the compartment; the occupancy condition includes a number of occupants and a minimum spacing between the number of occupants; determining the target configuration based on dimensional data of an object to be loaded into the compartment; generating, via the at least one time-of-flight sensor, a second point cloud representing the object to be loaded into the compartment, the second point cloud including three-dimensional positional information of the object, and calculating, via the processing circuitry, the dimensional data based on the second point cloud; the at least one time-of-flight sensor includes a first LiDAR module in the vehicle configured to capture the first point cloud and a second LiDAR module configured to capture the second point cloud; estimating a weight of the object based on the dimensional data, wherein determining the availability is based further on the weight of the object; identifying cargo in the compartment based on the first point cloud, and estimating a load distribution of the cargo, wherein determining the availability is based further on the load distribution; determining, via the processing circuitry, an unloading plan for the cargo based on the load distribution; determining an identity of an unloading source for the cargo based on the unloading plan, wherein the identity corresponds to a work capacity of the unloading source; the unloading plan includes route information for the vehicle and further includes determining the route information based on the load distribution; determining a timing for the unloading plan based on the load distribution; the timing is an arrival time for the vehicle along a target delivery route; and the vehicle is a first vehicle of a fleet of vehicles and further includes generating, via the at least one time-of-flight sensor, a second point cloud of a loading region of a second vehicle of the fleet of vehicles, the second point cloud including three-dimensional positional information of the compartment, comparing the first point cloud to the second point cloud, and determining a loading plan for the fleet of vehicles based on the comparison of the first point cloud to the second point cloud.

(5) According to a second aspect of the present disclosure, a system for determining spatial availability for a compartment of a vehicle. The system includes at least one time-of-flight sensor configured to generate a first point cloud representing the compartment of the vehicle. The first point cloud includes three-dimensional positional information of the compartment. The system further includes processing circuitry in communication with the at least one time-of-flight sensor configured to calculate spatial information corresponding to a target configuration for the compartment, compare the spatial information to the first point cloud, and determine an availability of the target configuration based on the comparison of the spatial information to the first point cloud.

(6) Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features: the at least one time-of-flight sensor includes a first LiDAR module in the vehicle configured to capture the first point cloud and a second LiDAR module configured to capture a second point cloud of an object to be loaded into the compartment, the second point cloud includes three-dimensional positional information of the object; the processing circuitry is further configured to determine the target configuration based on an occupancy condition of the compartment; and the occupancy condition includes a number of occupants and a minimum spacing between the number of occupants.

(7) According to a third aspect of the present disclosure, a method for determining spatial availability for a compartment of a vehicle. The method includes generating, via at least one LiDAR module, a first point cloud representing an interior of the vehicle, the first point cloud including three-dimensional positional information of the interior. The method further includes generating, via the at least one LiDAR module, a second point cloud representing an object to be loaded into the compartment of the vehicle. The second point cloud includes three-dimensional positional information of the object. The method further includes calculating, via processing circuitry in communication with the at least one LiDAR module, dimensional data of the object to be loaded based on the second point cloud. The method further includes determining, via the processing circuitry, a target configuration based on the dimensional data of the object to be loaded. The method further includes calculating, via the processing circuitry, spatial information corresponding to the target configuration for the compartment. The method further includes comparing, via the processing circuitry, the spatial information to the first point cloud. The method further includes determining an availability of the target configuration based on the comparison of the spatial information to the first point cloud.

(8) Embodiments of the third aspect of the present disclosure can include any one or a combination of the following features: the at least one LiDAR module includes a first LiDAR module in the vehicle configured to capture the first point cloud and a second LiDAR module configured to capture the second point cloud.

(9) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1A is a perspective view of a cargo van incorporating a detection system of the present disclosure in a rear space of the cargo van;

(3) FIG. 1B is a perspective view of a car incorporating a detection system of the present disclosure in a passenger cabin of the car;

(4) FIG. 2A is a representation of a point cloud generated by a time-of-flight sensor configured to monitor a rear space of a cargo van of the present disclosure;

(5) FIG. 2B is a representation of a point cloud generated by a time-of-flight sensor configured to monitor a passenger compartment of a vehicle of the present disclosure;

(6) FIG. 3 is a block diagram of an exemplary detection system incorporating light detection and ranging;

(7) FIG. 4 is a block diagram of an exemplary detection system for a vehicle;

(8) FIG. 5 is a perspective view of a loading scene for a compartment of a vehicle and an adjacent region exterior to the vehicle each being scanned by at least one LiDAR module of a detection system of the present disclosure;

(9) FIG. 6 is a side perspective view of a cargo van and removable seating being scanned by at least one LiDAR module demonstrating dynamic morphing for occupancy configuration modification;

(10) FIG. 7 is a side perspective view of a cargo van having a cutaway portion demonstrating cargo in the cargo van being monitored by a detection system of the present disclosure;

(11) FIG. 8 demonstrates a static morphing of a truck-bed to a harvester arrangement using a detection system of the present disclosure;

(12) FIG. 9 demonstrates a static morphing of a passenger-cargo compartment to a kitchen/living area arrangement using a detection system of the present disclosure;

(13) FIG. 10 is a block diagram of a detection system according to one aspect of the present disclosure;

(14) FIG. 11 is a functional diagram of a fleet management system operating as a global detection system of the present disclosure;

(15) FIG. 12 is a method of determining spatial availability for a compartment of a vehicle according to one aspect of the present disclosure; and

(16) FIG. 13 is a block diagram of an exemplary process for optimizing space morphing for a compartment of a vehicle.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(17) Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements may or may not be to scale and certain components may or may not be enlarged relative to the other components for purposes of emphasis and understanding.

(18) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1A. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(19) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to vehicle space morphing. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

(20) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(21) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

(22) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other,

such as within about 5% of each other, or within about 2% of each other.

(23) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

(24) Referring generally to FIGS. 1A-5, the present disclosure generally relates to a detection system **10** for a vehicle **12** that utilizes three-dimensional image sensing to detect information about an environment **14** in or around the vehicle **12**. The three-dimensional image sensing may be accomplished via one or more time-of-flight (ToF) sensors **16** that are configured to map a three-dimensional space such as an interior **18** of the vehicle **12** and/or a region exterior **20** to the vehicle **12**. For example, the one or more time-of-flight sensors **16** may include at least one light detection and ranging (LiDAR) module **22** configured to output pulses of light, measure a time of flight for the pulses of light to return from the environment **14** to the at least one LiDAR module **22**, and generate at least one point cloud **24** of the environment **14** based on the time-of-flight of the pulses of light. In this way, the LiDAR module **22** may provide information regarding three-dimensional shapes of the environment **14** being scanned, including geometries, proportions, or other measurement information related to the environment **14** and/or occupants **26** for the vehicle **12**.

(25) The LiDAR modules **22** of the present disclosure may operate conceptually similarly to a still frame or video stream, but instead of producing a flat image with contrast and color, the LiDAR module **22** may provide information regarding three-dimensional shapes of the environment **14** being scanned. Using time-of-flight, the LiDAR modules **22** are configured to measure the round-trip time taken for light to be transmitted, reflected from a surface, and received at a sensor near the transmission source. The light transmitted may be a laser pulse. The light may be sent and received millions of times per second at various angles to produce a matrix of the reflected light points. The result is a single measurement point for each transmission and reflection representing distance and a coordinate for each measurement point. When the LiDAR module **22** scans the entire “frame,” or field of view **30**, it generates an output known as a point cloud **24** that is a 3D representation of the features scanned.

(26) In some examples, the LiDAR modules **22** of the present disclosure may be configured to capture the at least one point cloud **24** independent of visible-light illumination of the environment **14**. For example, the LiDAR modules **22** may not require ambient light to achieve the spatial mapping techniques of the present disclosure. For example, the LiDAR module **22** may emit and receive IR or near-infrared (NIR) light, and therefore generate the at least one point cloud **24** despite visible-light conditions. Further, as compared to Radio Detection and Ranging (RADAR), the depth-mapping achieved by the LiDAR modules **22** may have greater accuracy due to the rate at which the LiDAR pulses may be emitted and received (e.g., the speed of light). Further, the three-dimensional mapping may be achieved without utilizing radio frequencies (RF), and therefore may limit or eliminate need for RF certifications for operation. Accordingly, sensors incorporated for monitoring frequencies and magnitudes of RF fields may be omitted by providing the present LiDAR modules **22**.

(27) Referring now more particularly to FIGS. 1A and 1B, a plurality of the LiDAR modules **22** may be configured to monitor a compartment **28** of the vehicle **12**. In the example illustrated in FIG. 1A, the LiDAR modules **22** are configured with a field of view **30** that covers the rear space of the vehicle **12**, as well as the region exterior **20** to the vehicle **12**. In this example, the region exterior **20** to the vehicle **12** is a space behind the vehicle **12** adjacent to an entry or an exit to the vehicle **12**. In FIG. 1B, the plurality of LiDAR modules **22** are configured to monitor a front space of the vehicle **12**, with the field of view **30** of one or more of the plurality of LiDAR modules **22** covering a passenger cabin **32** of the vehicle **12**. As will be described further herein, it is contemplated that the plurality of LiDAR modules **22** may be in communication with one another to allow the at least one point cloud **24** captured from each LiDAR module **22** to be compared to

one another to render a greater-accuracy representation of the environment **14**. For example, and as depicted in FIG. **1A**, the occupant **26** or another user may direct a mobile device **35** toward the environment **14** to generate an additional point cloud **24** from a viewing angle different than the field-of-views **30** of the LiDAR modules **22** of the vehicle **12**. For example, the mobile device **35** may be a cellular phone having one of the LiDAR modules **22**. In general, the time-of-flight sensors **16** disclosed herein may capture point clouds **24** of various features of the environment **14**, such as seats **34**, occupants **26**, and various other surfaces or items present in the interior **18** or the region exterior **20** to the vehicle **12**. As will further be discussed herein, the present system **10** may be operable to identify these features based on the at least one point cloud **24** and make determinations and/or calculations based on the identities, spatio-temporal positions of the features, and/or other related aspects of the features detected in the at least one point cloud **24**.

(28) Referring now to FIGS. **2A** and **2B**, representations of at least one point cloud **24** generated from the LiDAR modules **22** in the interiors **18** of the vehicles **12** of FIGS. **1A** and **1B**, respectively, are presented to illustrate the three-dimensional mapping of the present system **10**. For example, the depictions of the at least one point cloud **24** may be considered three-dimensional images constructed by the LiDAR modules **22** and/or processors in communication with the LiDAR modules **22**. Although the depictions of the at least one point clouds **24** illustrated in FIGS. **2A** and **2B** may differ in appearance, it is contemplated that such difference may be a result of averaging depths of the points **36** of each point cloud **24** to render a surface (FIG. **2B**) as opposed to individual dots (FIG. **2A**). The underlying 3D data may be generated the same way in either case.

(29) Still referring to FIGS. **2A** and **2B**, each point cloud **24** includes the three-dimensional data (e.g., a three-dimensional location relative to the LiDAR module **22**) for the various features in the interior **18**. For example, the at least one point cloud **24** may generate 3D mapping of the occupants **26** or cargo **37** in the interior **18**. The three-dimensional data may include the rectilinear coordinates, with XYZ coordinates, of various points **36** on surfaces or other light-reflective features relative to the LiDAR module **22**. It is contemplated that the coordinates of each point **36** may be virtually mapped to an origin point other than the LiDAR module **22**, such as a center of mass of the vehicle, a center of volume of the compartment **28** being monitored, or any other feasible origin point. By obtaining the three-dimensional data of the various features in the interior **18** and, in some cases, the region exterior **20** to the vehicle **12**, the present system **10** may provide for enhanced monitoring methods to be performed without complex imaging methods, such as those incorporating stereoscopic imagers or other three-dimensional monitoring devices that may require higher computational power or decreased efficiencies.

(30) Referring now to FIG. **3**, at least a portion of the present detection system **10** is exemplarily applied to a target surface **38**, such as to the cargo **37** or other surfaces in the environment **14** of the vehicle **12**. The system **10** may include processing circuitry **40**, which will be further discussed in relation to the proceeding figures, in communication with one or more of the time-of-flight sensors **16**. In the present example, the time-of-flight sensors **16** include the LiDAR modules **22** each having a light source **42**, or emitter, and a sensor **46** configured to detect reflection of the light emitted by the light source **42** off of the target surface **38**. A controller **48** of the LiDAR module **22** is in communication with the light source **42** and the sensor **46** and is configured to monitor the time-of-flight of the light pulses emitted by the light source **42** and returned to the sensor **46**. The controller **48** is also in communication with a power supply **50** configured to provide electrical power to the controller **48**, the light source **42**, the sensor **46**, and a motor **52** that is controlled by the controller **48**. In the present example, the LiDAR module **22** incorporates optics **54** that are mechanically linked to the motor **52** and are configured to guide the light pulses in a particular direction. For example, the optics **54** may include lenses or mirrors that are configured to change an angle of emission for the light pulses and/or return the light pulses to the sensor **46**. For instance, the motor **52** may be configured to rotate a mirror to cause light emitted from the light source **42** to

reflect off of the mirror at different angles depending on the rotational position of the motor 52.

(31) In some examples, the optics 54 may include a first portion associated with the source 42 and a second portion associated with the sensor 46. For example, a first lens, which may move in response to the motor 52, may be configured to guide (e.g., collimate, focus) the light emitted by the source 42, and a second lens, which may be driven by a different motor or a different connection to the motor 52, may be configured to guide the light reflected off the target surface 38 and returned to the sensor 46. Accordingly, the general configuration of the LiDAR module 22 may incorporate a single housing having different sets of optics or a plurality of housings with different optics. For example, the source 42 may be located in a first housing of the LiDAR module 22, the sensor 46 may be located in a second housing separate from or spaced from the first housing. In this way, each of the LiDAR modules 22 may refer to any emitter/receiver combination system that emits LiDAR pulses and receives the LiDAR pulses either at a common location in the vehicle 12 or at different locations in the vehicle 12.

(32) The light emitted and received by the present LiDAR modules 22 may have a wavelength in the range of between approximately 780 nanometers (nm) and 1700 nm. In some examples, the wavelength of the LiDAR is preferably in the range of between 900 nm and 1650 nm. In other examples, the wavelength of the LiDAR is preferably between 1500 nm and 1650 nm. In some examples, the wavelength of the LiDAR is preferably at least 1550 nm. It is contemplated that the particular wavelength/frequency employed by the LiDAR modules 22 may be based on an estimated distance range for capturing the depth information. For example, for shorter ranges (e.g., between 1 m and 5 m) the LiDAR may operate with a greater wavelength of light (e.g., greater than 1000 nm). The LiDAR modules 22 of the present disclosure may be configured to output light, in the form of a laser, at a wavelength of at least 1550 nm while the motor 52 rotates the optics 54 to allow mapping an area. In some examples, the LiDAR modules 22 of the present disclosure are configured to emit light having a wavelength of at least 1650 nm. Due to the relatively short distances scanned by the present LiDAR modules 22 (e.g., between one and five meters), such relatively low infrared (IR) or near-infrared (NIR) may be employed to achieve the three-dimensional spatial mapping via the at least one point cloud 24 with low power requirements. The present LiDAR modules 22 may be either single point-and-reflect modules or may operate in a rotational mode, as described above. In rotational mode, the LiDAR module 22 may measure up to 360 degrees based on the rate of rotation, which may be between 1 and 100 Hertz or may be at least 60 rotations per minute (RPM) in some examples.

(33) In the example depicted in FIG. 3, the time-of-flight for a first pulse of light 56 emitted by the light source 42 and returned to the sensor 46 may be less than a second time-of-flight for a second light pulse emitted by the light source 42 returned to the sensor 46. For example, the first pulse of light 56 may travel a shorter distance than the second pulse of light 58 due to a difference in depth, height, or width of the corresponding reflection point 36 on the target surface 38. In this way, the LiDAR module 22 may generate the at least one point cloud 24 to be representative of the environment 14 (e.g., the target surface 38 in the present example) in three dimensions.

(34) The processing circuitry 40 of the present disclosure may be provided to amalgamate the point cloud 24 from each of a plurality of the LiDAR modules 22 and process the coordinates of the features to determine an identity of the features, as well as to perform other processing techniques that will be further described herein. The processing circuitry 40 may include a first processor 40a local to the vehicle 12 and a second processor 40b remote from the vehicle 12. Further, the processing circuitry 40 may include the controller 48 of the LiDAR module 22. In some examples, the controller 48 may be configured to generate or determine the at least one point cloud 24 and/or point cloud data, and the first processor 40a may be configured to receive the at least one point cloud 24 from each LiDAR module 22 and compile each point cloud 24 of a common scene, such as the environment 14, to generate a more expansive or more accurate point cloud 24 of the environment 14.

(35) The second processor **40b**, which may be a part of a remote server **60** and in communication with the first processor **40a**, via a network **62**, may be configured to perform various modifications and/or mapping of the at least one point cloud **24** to target three-dimensional image data for the environment **14**. For example, the server **60** may include an artificial intelligence (AI) engine **64** configured to train machine learning models **66** based on the point cloud data captured via the LiDAR modules **22** and/or historical data previously captured by the time-of-flight sensors **16**. The second processor **40b** may be in communication with the AI engine **64**, as well as in communication with a database **67** configured to store the target point cloud data and/or three-dimensional image information. Accordingly, the server **60** may incorporate a memory storing instructions that, when executed by the processor, cause the processing circuitry **40** to compare the at least one point cloud **24** to point cloud data corresponding to target conditions of the interior **18** and/or the region exterior **20** to the vehicle **12**. In this way, the detection system **10** may employ the processing circuitry **40** to perform advanced detection techniques and to communicate with subsystems of the vehicle **12**, as will be described in the proceeding figures. In this way, the detection system **10** may be employed in tandem or in conjunction with other operational parameters for the vehicle **12**. For example, the detection system **10** may be configured for communicating notifications to the occupants **26** of alert conditions, controlling the various operational parameters in response to actions detected in the interior **18**, activating or deactivating various subsystems of the vehicle **12**, or interacting with any vehicle systems to effectuate operational adjustments.

(36) Referring now to FIG. 4, the detection system **10** may incorporate or be in communication with various systems of the vehicle **12** (e.g., vehicle systems). For example, the processing circuitry **40** may be configured to communicate with an imaging system **68** that includes imaging devices, such as cameras (e.g., red-, green-, and blue-pixel (RGB) or IR cameras). The processing circuitry **40** may further be in communication with other vehicle systems, such as a door control system **69**, a window control system **70**, a seat control system **71**, a climate control system **72**, a user interface **74**, mirrors **76**, a lighting system **78**, a restraint control system **80**, a powertrain **82**, a power management system **83**, or any other vehicle systems. Communication with the various vehicle systems may allow the processing circuitry **40** to transmit and receive signals or instructions to the various vehicle systems based on processing of the at least one point cloud **24** captured by the time-of-flight sensors **16**. For example, when the processing circuitry **40** identifies a number of occupants **26** in the vehicle **12** based on the at least one point cloud **24**, the processing circuitry **40** may communicate an instruction to adjust the seat control system **71** and/or the climate control system **72**. In another non-limiting example, the processing circuitry **40** may receive information or signals from the lighting system **78** and control operation of the time-of-flight sensors **16** based on the information from the lighting system **78**. Accordingly, the processing circuitry **40** may control, or communicate instructions to control, the time-of-flight sensors **16** based on information from the vehicle systems and/or may communicate signals or instructions to the various vehicle systems based on information received from the time-of-flight sensors **16**.

(37) The window control system **70** may include a window motor **84** for controlling a position of a window of the vehicle **12**. Further, the window control system **70** may include dimming circuitry **86**, which may be glazing dimming circuitry **86**, for controlling an opacity and/or level of light transmitted between the interior **18** of the vehicle **12** and the region exterior **20** to the vehicle **12**. One or more sunroof motors **88** may be provided with the window control system **70** for controlling closing and opening of a sunroof panel. It is contemplated that other devices may be included in the window control system **70**, such as window locks, window breakage detection sensors, and other features related to operation of the windows of the vehicle **12**. By providing communication between the window control system **70** and processing circuitry **40** of the present disclosure, the window control system **70** may be configured to adjust one or more of its features based on conditions determined or detected by the processing circuitry **40** based on the at least one

point cloud **24**. Similarly, the window control system **70** may transmit one or more signals to the processing circuitry **40**, and the processing circuitry **40** may control operation of the time-of-flight sensors **16** based on the signals from the window control system **70**.

(38) The climate control system **72** may include one or more heating and cooling devices, as well as vents configured to distribute heated or cooled air into the interior **18** of the vehicle **12**. Although not specifically enumerated in FIG. **4**, the climate control system **72** may be configured to actuate a vent to selectively limit and allow heated air or cooled air to circulate in the interior **18** of the vehicle **12**. Further, the climate control system **72** may be configured to operate heating, ventilation, and air conditioning (HVAC) systems to recirculate air or to vent air to the region exterior **20** to the vehicle **12**.

(39) The seat control system **71** may include various positioning actuators **90**, inflatable bladders **92**, seat warmers **94**, and/or other ergonomic and/or comfort features for seats **34** in the vehicle **12**. For example, the seat control system **71** may include motors configured to actuate the seat **34** forward, backward, up, down, side to side, or rotationally. Both a backrest of the seat **34** and a lower portion of the seat **34** may be configured to be adjusted by the positioning actuators **90**. The inflatable bladders **92** may be provided within the seat **34** to adjust a firmness or softness of the seat **34**, and seat warmers **94** may be provided for warming cushions in the seat **34** for comfort of the occupants **26**. In one non-limiting example, the processing circuitry **40** may compare the position of the seats **34** based on seat sensors **95**, such as position sensors, occupancy detection sensors, or other sensors configured to monitor the seats **34**, to the point cloud data captured by the time-of-flight sensors **16** in order to verify or check an estimated seat position based on the point cloud data. In other examples, the processing circuitry **40** may communicate one or more signals to the seat control system **71** based on body pose data identified in the at least one point cloud **24**. In yet further examples, the processing circuitry **40** may be configured to adjust an operational parameter of the time-of-flight sensors **16**, such as a scanning direction, a frequency of the LiDAR module **22**, or the like, based on the position of the seats **34** being monitored by the time-of-flight sensors **16**.

(40) The user interface **74** may include a human-machine interface (HMI) **96** and/or may include audio devices, such as microphones and/or speakers, mechanical actuators, such as knobs, buttons, switches, and/or a touchscreen **98** incorporated with the HMI **96**. The human-machine interface **96** may be configured to present various digital objects representing buttons for selection by the user via, for example, the touchscreen **98**. In general, the user interface **74** may communicate with the processing circuitry **40** to activate or deactivate the time-of-flight sensors **16**, adjust operational parameters of the time-of-flight sensors **16**, or control other aspects of the time-of-flight sensors **16**. Similarly, the processing circuitry **40** may be configured to communicate instructions to the user interface **74** to present information and/or other data related to the detection and/or processing of the at least one point cloud **24** based on the time-of-flight sensors **16**. It is further contemplated that the mobile device **35** may incorporate a user interface **74** to present similar options to the user at the mobile device **35**.

(41) Still referring to FIG. **4**, other vehicle systems include the mirrors **76**, the lighting system **78**, and the restraint control system **80**. These other vehicle systems may also be adjusted based on the at least one point cloud **24** generated by the time-of-flight sensors **16** and processed by the processing circuitry **40**. Additionally, subcomponents of these systems (e.g., sensors, processors) may be configured to send instructions or data to the processing circuitry **40** to cause the processing circuitry **40** to operate the time-of-flight sensors **16** in an adjusted operation. For example, the processing circuitry **40** may be configured to deactivate the time-of-flight sensors **16** in response to the lighting system **78** detecting adequate lighting to allow for visible light and/or IR occupant monitoring. In some examples, the processing circuitry **40** may communicate an instruction to adjust a position of the mirrors **76** based on the at least one point cloud **24**. For example, the at least one point cloud **24** may demonstrate an event, such as an orientation of a driver, a position of

another vehicle in the region exterior **20** to the vehicle **12**, or any other positional feature, and generate a signal to the mirrors **76** (or associated positioning members) to move the mirrors **76** to align a view with the event.

(42) Referring again to FIG. **4**, the vehicle **12** may include the powertrain **82** that incorporates an ignition system **100**, a steering system **102**, a transmission system **104**, a brake system **106**, and/or any other system configured to drive the motion of the vehicle **12**. In some examples, the at least one point cloud **24** captured by the time-of-flight sensors **16** may be processed by the processing circuitry **40** to determine target steering angles, rates of motion or speed changes, or other vehicle operations for the powertrain **82**, and communicate the target operations to the powertrain **82** to allow for at least partially autonomous control over the motion of the vehicle **12**. Such at least partially autonomous control may include fully autonomous operation or semiautonomous operation of the vehicle **12**. For example, the processing circuitry **40** may communicate signals to adjust the brake system **106**, the ignition system **100**, the transmission system **104**, or another system of the powertrain **82** to stop the vehicle **12** or move the vehicle **12**.

(43) The processing circuitry **40** may further include an occupant monitoring module **108** that may communicate with any of the vehicle systems described above, as well as the time-of-flight sensors **16** of the present disclosure. The occupant monitoring module **108** may be configured to store various algorithms for detecting aspects related to the occupants **26**. For example, the algorithms may be executed to monitor the interior **18** of the vehicle **12** to identify occupants **26** in the vehicle **12**, a number of occupants **26**, or other occupancy features of the interior **18** using the point cloud data and/or video or image data captured by the imaging system **68**. Similarly, various seat sensors **95** of the seat control system **71**, heating or cooling sensors that detect manual manipulation of the vents for heating or cooling control for the climate control system **72**, inputs to the window control system **70**, or any other sensor of the vehicle systems previously described may be processed in the occupant monitoring module **108** to detect positions of occupants **26** in the vehicle **12**, conditions of occupants **26** in the vehicle **12**, states of occupants **26** in the vehicle **12**, or any other relevant occupancy features that will be described herein. The processing circuitry **40** may also include various classification algorithms for classifying objects detected in the interior **18**, such as for the cargo **37**, mobile devices **35**, animals, and any other living or nonliving item in the interior **18**. Accordingly, the processing circuitry **40** may be configured to identify an event in the interior **18** or predict an event based on monitoring of the interior **18** by utilizing information from the other vehicle systems.

(44) In general, the detection system **10** may provide for spatial mapping of the environment **14** of the vehicle **12**. For example, the LiDAR modules **22** may detect the position, in three-dimensional space, of objects, items, or other features in the interior **18** or the region exterior **20** to the vehicle **12**. Such positions, therefore, include depth information of the scene captured by the LiDAR module **22**. As compared to a two-dimensional image captured by a camera, the at least one point cloud **24** generated by the time-of-flight sensor **16** allows for more efficient determination of how far the features are from the LiDAR module **22** and from one another. Thus, complex image analysis techniques involving pixel analysis, comparisons of RGB values, or other techniques to estimate depth may be omitted due to utilization of the ToF sensors **16**. Further, while multiple imaging devices from different angles of a common scene (e.g., a stereoscopic imager) may allow for more accurate estimation of depth information than those produced by a single camera, complex data processing techniques may be required for multiple cameras to be employed to gather the depth information. Further, such multi-camera systems may require additional weight, packaging volume, or other inefficiencies relative to the time-of-flight sensors **16** of the present disclosure.

(45) Accordingly, the detection system **10** may be computationally-efficient and/or power-efficient relative to two-dimensional and three-dimensional cameras for determining positional information. Further, other time-of-flight sensing techniques, such as RADAR, while providing depth

information, may present certification issues based on RF requirements and may be less accurate than the present LiDAR modules **22**. Further, a number of cameras used for monitoring the environment **14** may be reduced, various presence detectors (vehicle seat sensors **95**) may be omitted, and other sensors configured to determine positional information about the environment **14** may be omitted due to the precision of the LiDAR. Thus, a solution may be provided by the detection system **10** by reducing the number of sensors required to monitor various aspects of the environment **14**.

(46) Referring to FIGS. 5-13, the present system **10** may be configured to determine spatial availability for the compartment **28** of the vehicle **12**. The system **10** includes at least one of the time-of-flight sensors **16**. The time-of-flight sensor **16** is configured to generate a first point cloud **24a** representing the compartment **28** of the vehicle **12**. The first point cloud **24a** includes three-dimensional positional information of the compartment **28** of the vehicle **12**. The processing circuitry **40** is in communication with the at least one time-of-flight sensor **16** and is configured to calculate spatial information corresponding to a target configuration for the compartment **28**. The processing circuitry **40** is further configured to compare the spatial information to the first point cloud **24a**. The processing circuitry **40** is further configured to determine an availability of the target configuration based on the comparison of the spatial information to the first point cloud **24a**. The availability may refer to the feasibility or operability of the target configuration to be applied to the compartment **28**. For example, as will be described further herein, the target configuration may refer to a static, a pseudo-static, or a dynamic morphing level for the vehicle **12**, and the processing circuitry **40** may be configured to determine whether such target configurations are possible given the spatial information and the first point cloud **24a**. In some examples, the at least one time-of-flight sensor **16** includes a first LiDAR module **22a** configured to generate the first point cloud **24a** of the compartment **28** and a second LiDAR module **22b** configured to generate a second point cloud **24b** of objects **138** to be loaded.

(47) Static morphing for the vehicle **12** may refer to permanent changes and/or up-fits that may result in modification of structural components for the vehicle **12**, such as modification to mounts for the seating, modification of interior walls **120** of the vehicle **12**, or the like. Generally, static morphing may refer to a significant change in the function of the interior **18** of the vehicle **12** or in another compartment **28** of the vehicle **12** (e.g., bed **160** of a truck **162** converted to a cherry picker). Pseudo-static morphing may refer to cyclical updates and arrangements that may be modified but may require significant time and energy to change to or from a particular configuration. For example, a van **166** may be pseudo-statically morphed to include an office area in lieu of a cargo area in a rear of the van **166**. Dynamic morphing may refer to spatial changes based on unloading or loading, picking up or dropping off cargo **37**, swapping seating configurations by removing or selectively deploying particular seats **34** in the vehicle **12**, adding a child seat and/or removing a child seat, or the like. In general, the differences between the morphing levels may be defined based on the amount of time or energy required to adjust an existing configuration to a target configuration and/or a degree of modification to structural components of the compartment **28**. For example, as previously described, the static morphing level may involve significant modifications to the structures of the interior **18** to allow for new functions, dynamic morphing may refer to nonstructural modifications to the interior **18**, and pseudo-static morphing may refer to slight or limited modifications to structural components of the interior **18**.

(48) In general, the present system **10** may be configured to utilize the time-of-flight sensors **16** to produce a three-dimensional map of the interior **18** and overlay the target configuration to determine whether such target configuration is feasible based on spatial limitations of the vehicle **12**. As will be described further herein, the system **10** may be configured to access Computer-aided design (CAD) data **122** (FIG. **10**) or reference data based on the individual model of the vehicle **12** and/or CAD data **122** or reference data of information relevant to objects **138** stored in the interior

18. For example, information related to packages may be stored in a database **67** after being scanned for a delivery service. Such information may include the size, weight, or other features related to three-dimensional information of an item. Further, the time-of-flight sensors **16** may be employed to capture occupancy configurations and/or seating configurations to allow for recommendation of specific dynamic morphing configurations, such as seating rearrangements or the like depending on the number of occupants **26**, a comfort level for the occupants **26**, or other user-driven preferences.

(49) Referring now more particularly to FIG. 5, an example of a dynamic morphing level selection process is generally illustrated with the present time-of-flight sensors **16** employed to scan the interior **18** as well as the region exterior **20** to the vehicle **12** in order to generate a first point cloud **24a** of the interior **18** and the second point cloud **24b** of the region exterior **20**. The first point cloud **24** may represent the cargo **37** in the interior **18**. The time-of-flight sensors **16** may be employed to generate the second point cloud **24b** for the objects **138** in the region exterior **20** to the vehicle **12**. As will be described further in relation to the proceeding figures, the vehicle **12** may include one or more weight sensors **124** positioned in the vehicle **12** for monitoring a load distribution in the vehicle **12**. For example, the weight sensors **124** may be placed, as illustrated, at corners **126** of the rear compartment of the vehicle **12**, though any other placement of the weight sensors **124** that allows for determining a load distribution in the vehicle **12** may be provided. As depicted in FIG. 5, a user may manipulate the mobile device **35**, such as a phone, to adjust a time-of-flight sensor **16** of the mobile device **35**. For example, one LiDAR module **22** may be incorporated into the mobile device **35**, and three-dimensional positional information captured by the LiDAR module **22** of the mobile device **35** may be communicated to the processing circuitry **40**, via the network **62** previously described. In this way, remote devices configured to capture the at least one point cloud **24** from different angles may be employed for greater accuracy in determining the availability for the target configuration.

(50) Still referring to FIG. 5, the objects **138** in the region exterior **20** may be intended to be loaded into the interior **18**. Such objects **138** may include boxes **128** on a pallet **130** as well as other atypically-shaped packages **132** or other items located in a loading region for the vehicle **12**. In the present example, the first point cloud **24a** includes the three-dimensional positional information of the walls **120**, the floor **134**, the ceiling **136**, and the cargo **37** already in the interior **18**, and the processing circuitry **40** calculates spatial information for a target configuration in which the objects **138** in the region exterior **20** to the vehicle **12** are disposed in the interior **18** alongside objects **138** already in the interior **18**. Thus, the processing circuitry **40** may calculate any number of target configurations that may be physically possible to fit in the interior **18**. Because the time-of-flight sensors **16** employed may be configured to map the depth of the objects **138**, including the atypically-shaped package **132**, the detection system **10** may continuously monitor the interior **18** while the interior **18** is being loaded to limit the number of possible target configurations that may be implemented. For example, if the atypically-shaped package **132** is loaded initially and placed toward the rear of the interior **18**, the number of potential target configurations may be decreased, and different loading recommendations may be determined by the processing circuitry **40**. It is contemplated that, due to the precision and quality of the points **36** generated by the LiDAR modules **22**, the processing circuitry **40** may be configured to estimate the volume or space availability with greater accuracy than estimates based on other time-of-flight sensors **16** (e.g., RADAR) or cameras.

(51) In general, the processing circuitry **40** may determine the target configuration of the compartment **28**. For example, the occupancy condition may, in addition or in an alternative, include a minimum spacing between occupants **26** based on health guidelines. In some examples, the occupancy condition includes a number of occupants in the vehicle **12**.

(52) Still referring to FIG. 5, the time-of-flight sensors **16** may further be configured to generate the at least one point cloud **24** representing a loading source **140** and/or an unloading source **140**. For

example, the processing circuitry **40** may be configured to determine an identity of the loading or the unloading source **140** for the cargo **37** and/or the objects **138**. As will be described further herein, the processing circuitry **40** may utilize the identity of the loading or the unloading source **140** to calculate an unloading plan for the cargo **37** or a loading plan for the objects **138**. For example, the identity of the loading/unloading source **140** may correspond to a human **142**, such as the user, or a machine **144**, such as a forklift **146** or other mechanical or electromechanical machine configured to aid in lifting or loading objects **138** into or out of the compartment **28**. Thus, the detection system **10** of the present disclosure may utilize the time-of-flight sensors **16** to estimate loading times, unloading times, recommended target configurations (e.g., an order of the objects **138** in the interior **18**), or various other loading or unloading features of the present disclosure.

(53) Referring now to FIGS. **6** and **7**, other examples of dynamic morphing are illustrated in reference to occupancy configurations (FIG. **6**) and load distributions (FIG. **7**). For example, the detection system **10** may be configured to detect a rearrangement of seats **34** based on the at least one point cloud **24** of the interior **18** (e.g., the first point cloud **24a**) and/or the at least one point cloud **24** of the region exterior **20** to the vehicle **12** (e.g., the second point cloud **24b**). For example, the second point cloud **24b** may represent a removed seat **148** from the vehicle **12** and/or a booster seat **150** removed from the vehicle **12**. Based on historical data (e.g., previous LiDAR scans), the processing circuitry **40** may determine that the removed seat **148** and the booster seat **150** have been removed and/or are being placed in the vehicle **12**. Accordingly, the processing circuitry **40** may detect the updated intended seating configuration (e.g., the target configuration) and present a plurality of options **182** for the users to choose from based on the automatic detection of a new intended target configuration. As will be described further herein, the recommended target configurations may be based on occupancy preferences, such as spacing from the booster seat **150** and/or spacing amongst the occupants **26** based on health guidelines (e.g., social distancing).

(54) With specific reference to FIG. **7**, the detection system **10** may utilize the time-of-flight sensors **16** in combination with the weight sensors **124** in the vehicle **12** to determine the load distribution within the interior **18**. For example, the processing circuitry **40** may identify the spatial availability in the interior **18** and map portions of the spatial availability. For example, in a first region **152** of the vehicle **12**, a stack **154** of cargo **37** may exceed or approach a recommended stack height **156**, and in a second region **158**, the stack height **156** may be below the recommended stack height **156**. The spatial availability may be compared to the weight distribution, or load distribution within the interior **18** to determine recommended placement of new objects **138** in the vehicle **12** in spite of the spatial availability. For example, if the second region **158** includes more spatial availability than the first region **152**, but the load is distributed toward the second region **158** (e.g., the objects **138** in the second region **158** are heavier than the objects **138** in the first region **152**), the detection system **10** may determine that no free space is actually available for new objects **138**. In this way, the processing circuitry **40** may utilize the information gathered from the weight sensors **124** and the time-of-flight sensors **16** to optimize loading and/or unloading procedures for the vehicle **12**.

(55) As will be also described further herein, the loading/unloading plans and/or navigational routes for the vehicle **12** may be determined based on the spatial availability and/or the target configuration for the vehicle **12**. For example, if a plurality of potential objects **138** may be desired to be loaded into the vehicle **12**, the processing circuitry **40** may access information about the objects **138** (e.g., a packaging weight, height, width, depth, or other spatial parameter) and determine whether or which of the plurality of objects **138** may be capable of loading into the vehicle **12**. For example, the packaging parameters may be scanned during distribution (via, e.g., a barcode or RFID scanner) and uploaded to a digital library accessible by the processing circuitry **40**. Based on the recommendation calculated by the processing circuitry **40**, the navigational route may be adjusted to pick up or to not pick up certain packages along a target path.

(56) Referring now to FIGS. **8** and **9**, two exemplary static morphings are illustrated in the form of

a modification of a bed **160** of a truck **162** to operate as agricultural equipment **164** (FIG. **8**) and as a modification of the passenger cabin **32** of a van **166** to serve as a kitchen and living area **168** for the van **166** (FIG. **9**). With particular reference to FIG. **8**, the detection system **10** may employ the time-of-flight sensors **16** to generate the at least one point cloud **24** of the bed **160** of the truck **162** and/or access specification data, such as CAD data **122** that includes dimensional information of the bed **160** of the truck **162**, to generate a three-dimensional mapping of the bed **160** of the truck **162**. If a user desires the agricultural function illustrated in FIG. **8** or any other static or permanent modification to the vehicle **12**, the user may, via the user interface **74**, enter a request to modify the truck **162** to the target configuration incorporating the agricultural function. Based on the three-dimensional mapping of the bed **160** of the truck **162** and the spatial information corresponding to the target configuration (e.g., a volume, and/or a geometry required for the static morphing), the processing circuitry **40** may determine one or more possible configurations for the agricultural function. Accordingly, the processing circuitry **40** may limit the number of possible configurations or deny the possibility of such a target configuration. For example, the processing circuitry **40** may be configured to estimate a load distribution for the target configuration of the truck **162** and compare the load distribution to parameters required for operation of the vehicle **12**, such as maneuverability, engine power requirements, or any other vehicle operation parameter. In this way, the present system **10** may analyze exterior compartments for vehicles **12** and aid a user in selection of a permanent modification.

(57) With particular reference to FIG. **9**, the user has selected a kitchen/living area **168** modification to the passenger cabin **32** for the van **166**. The processing circuitry **40** may be configured to, based on depth information from the at least one point cloud **24** generated by the time-of-flight sensors **16**, recommend the target configuration illustrated and/or additional target configurations for the passenger cabin **32** based on the spatial mapping and the at least one point cloud **24**. It is also contemplated that the processing circuitry **40** may be configured to limit or select potential configurations for the target configuration based on other parameters entered by the user, such as space optimization, seating optimization, location of electrical power, water sources, or other user parameters that may be described further herein. For example, the kitchen/living area **168** selected by the processing circuitry **40** may limit such possible configurations to include a sink **170** of the kitchen area located toward a front of the vehicle **12** or another portion of the vehicle **12** where a water or electrical supply may be available. Continuing with this example, the lighting fixtures **172** in the kitchen/living area **168** selected by the processing circuitry **40** may be required to be near a particular electrical power supply and therefore limit lighting options within the interior **18**. In this way, the present system **10** may employ the present LiDAR detection techniques in combination with access to three-dimensional information of possible static morphing options for vehicle **12** to optimize selection of a target configuration.

(58) Referring now to FIG. **10**, the processing circuitry **40** may be in communication with the LiDAR modules **22** as well as the weight sensors **124** previously described. Further, the processing circuitry **40** may be in communication with the mobile device **35** or a plurality of mobile devices, which may employ the network **62** previously described to communicate with the processing circuitry **40**. CAD data **122** related to the dimensional information for the vehicle **12** may also be in communication with the processing circuitry **40**. For example, remote databases that will be further described with reference to FIG. **11** may include the CAD data **122** which may be compared to the target configuration and/or the at least one point cloud **24** generated by the LiDAR modules **22** in order to determine the availability of the target configuration. A navigational system **174**, which may be one of the vehicle systems previously described, may also be in communication with the processing circuitry **40**. The navigational system **174** may also be in communication with the mobile device **35** and/or include the mobile device **35** and be configured to predict and/or receive destination locations, initial locations, stopping points, or the like for suggesting routes between such locations. For example, the navigational system **174** may include a global positioning system

(GPS) **176** or have access to the internet in order to calculate routes, estimate arrival times, travel durations, or the like.

(59) Referring now to FIG. **11**, the detection system **10** may be utilized for operation with a fleet of vehicles **178** in order to determine target configurations for a fleet of vehicles **178**, or a plurality of vehicles **178**. For example, each of the fleet of vehicles **178** may incorporate the detection system **10**, and each detection system **10** may be in communication with one another to create a global fleet management system **10**. In this example, a loading plan for the fleet of vehicles **178** may be determined by the processing circuitry **40** based on a comparison of at least one point cloud **24** for each vehicle. As an example, a first vehicle and a second vehicle of the fleet of vehicles **178** may each have a time-of-flight sensor **16** that is configured to generate the at least one point cloud **24** of an environment **14** corresponding to the respective first and second vehicles. The processing circuitry **40** may then compare the at least one point cloud **24** captured in the environment **14** of the first vehicle to the at least one point cloud **24** of the environment **14** for the second vehicle, and determine the loading plan based on the comparison of those two point clouds **24**. It is contemplated that this process may be carried out recursively across the entirety of the fleet of vehicles **178** in order to provide for an optimized loading plan for the fleet of vehicles **178**.

(60) With continued reference to FIG. **11**, the processing circuitry **40** may be further configured to recommend particular enhancements or target configurations based on cargo parameters **180** that may be entered by the user via the user interface **74** or may be loaded automatically based on a bill of sale, packaging information scanned into the database **67** or another digital library, or the like. In other examples, the upfit recommendations may be input to the processing circuitry **40** by the user, at the user interface **74**, or may be provided as choices for a given set of cargo parameters **180**. For example, if 3 tons of cargo must be shipped to a desired location via the fleet of vehicles **178**, the processing circuitry **40** may access scans **184** of the vehicle **12** to determine the spatial and/or max cargo weight limitations of each of the fleet of vehicles **178**, and present to the user a plurality of options for morphing each of the vehicles **12** in the fleet of vehicles **178** in order to accomplish the 3-ton loading/unloading plan. As depicted, the recommendations may include dynamic, static, or pseudo-dynamic morphing levels to accomplish the desired loading or unloading plan. In this way, the fleet management system **10** may provide flexibility by determining the options **182** for performing the loading plan and/or limiting other options **182** from the loading plan based on the cargo parameters **180**. It is contemplated that, although previously described as an operation with the user interface **74**, the fleet management system **10** may include a fleet management hub **186** at which the user may interact with and/or select the target configurations for each of the plurality of vehicles **178** and/or may enter or select limitations related to, for example, spatial optimization, style, comfort, or the like.

(61) Referring now to FIG. **12**, a method **1200** for determining spatial availability for the compartment **28** of the vehicle **12** includes generating, via at least one of the time-of-flight sensors **16**, the first point cloud **24a** representing the compartment **28** of the vehicle **12** at step **1202**. The method further includes calculating, via the processing circuitry **40** in communication with the at least one time-of-flight sensor **16**, spatial information corresponding to a target configuration for the compartment **28** at step **1204**. The method **1200** further includes comparing, via the processing circuitry **40**, the spatial information to the first point cloud **24a** at step **1206**. The method **1200** further includes determining an availability of the target configuration based on the comparison of the spatial information to the first point cloud **24a** at step **1208**. It is contemplated that the compartment **28** may be in the interior **18** of the vehicle **12** or may be accessible from the region exterior **20** to the vehicle **12**.

(62) In some examples, generation of the first point cloud **24a** and/or the second point cloud **24b** may be accomplished via the controller **48** for the LiDAR module **22** previously described, the first processor **40a** previously described, and/or the second processor **40b** previously described. Thus, any portion of the processing circuitry **40** or another processor may generate the at least one point

cloud **24** based on the data captured by the LiDAR modules **22**. It is further contemplated that the spatial information calculated by the processing circuitry **40** may include volumetric data, mass information, geometrical information, proportional data, or any other metric related to positional information about the compartment **28**. It is further contemplated that the compartment **28** scanned by the time-of-flight sensor **16** may be the passenger cabin **32** previously described, the bed **160** of the truck **162**, another exterior compartment, a different compartment **28** in the interior **18**, such as the rear space of a cargo van **166**, or any other compartment.

(63) In other examples, the method **1200** may further include generating, via the at least one time-of-flight sensor **16**, the second point cloud **24b** of an object **138** to be loaded into the compartment **28**. The method **1200** further may include calculating, via the processing circuitry **40**, the dimensional data based on the second point cloud **24b**. The dimensional data may refer to the measurement information of the object **138** to be loaded, such as a height, a weight, a width, a length, a geometry, a shape, or any other positional aspect of the object **138**. It is further contemplated that the method **1200** may include determining the target configuration based on the dimensional data. Accordingly, and with reference back to FIG. 5, the pallet **130** of boxes **128** and/or the atypically-shaped package **132** may be the objects **138** to be loaded, and the dimensional data mapped via the LiDAR modules **22** may be fed to the processing circuitry **40**, which may utilize the dimensional data to calculate the spatial information for the target configuration.

(64) In some examples, the method **1200** further includes estimating a weight of the object **138** to be loaded based on the dimensional data. For example, the weight of the object **138** may not be otherwise known to the processing circuitry **40** and may be calculated based on estimations of density and/or volumetric information. In other examples, documentation related to bills of sales, inventories, or other information accessible by the processing circuitry **40** may include the packaging information, which may have the weight of the objects **138**, the sizes of the objects **138**, and/or other parameters. For example, the documentation may include an indication of a loading or unloading method (e.g., recommending a two-person lift, a single person left, a mechanically assisted lift process, etc.).

(65) In other examples, the method **1200** includes identifying the cargo **37** in the compartment **28** based on the at least one point cloud **24** and estimating the load distribution of the cargo **37** in the vehicle **12**. As previously described, the load distribution may relate to known weights or estimated weights of the objects **138**, and scanning via the LiDAR modules **22** may provide the location/relative location of the objects **138** in the compartment **28** and allow the processing circuitry **40** to determine the load distribution. In other examples, the weight sensors **124** may be employed to validate or compare the estimated load distribution by the processing circuitry **40**. Accordingly, the processing circuitry **40** may determine the unloading plan or the loading plan for the cargo **37** based on the load distribution calculated or estimated by the processing circuitry **40**. It is also contemplated that the power management system **83** may communicate an instruction to the processing circuitry **40** to optimize the load distribution to conserve electrical power for the vehicle **12**. For example, one or more batteries may be provided for the power management system **83**, the power management system **83** may communicate a signal to recommend or select a load distribution as the target configuration. In some examples, the processing circuitry **40** may detect sub-optimal load distributions based on the at least one point cloud **24** and/or weight distribution detected by the weight sensor and recommend an organization for the cargo that enhances load distribution resulting in lower power consumption due to, for example, limiting necessary braking power (for a hydraulic pump or electric motors) due to an evenly distributed load. In this way, power consumption may be limited due to the detection system **10**.

(66) As previously described with respect to FIG. 5, the identity of an unloading source **140** for the cargo **37** may be determined by the processing circuitry **40** based on the unloading plan. For example, if the processing circuitry **40** determines the unloading plan based on the load distribution, the unloading plan may include determining an identity of the unloading source **140**

(e.g., a human **142** or a mechanical assistance device **144**). The identity may correspond to a work capacity of the unloading source **140**. For example, the processing circuitry **40** may determine the unloading source **140** to be able to lift over 100 pounds, for example, or some other weight associated with the objects **138**, a size of the objects **138**, or another parameter of the objects **138**. For example, if the pallet **130** is aboard the vehicle **12**, the processor may determine that a Hi-Lo, a jack, a forklift **146**, or the like, should be used to unload the vehicle **12**. In other examples, the processing circuitry **40** may determine that the unloading source **140** is a human **142**, and may further recommend the human **142** have particular attributes related to sex, age, or other factors relating to lifting/endurance capacity.

(67) In other examples, the unloading plan includes the route information for the vehicle **12** previously described with respect to the navigational system **174**. The processing circuitry **40** may be further configured to determine the route information based on the load distribution. For example, if certain unloading sources **140** and/or loading sources **140** are available at particular locations along a travel route, the order of stops along the travel route may be selected by the processing circuitry **40** to allow for more efficient unloading or loading plans. Accordingly, the processing circuitry **40** may determine a timing for the unloading plan based on the load distribution. The timing may be an arrival time, a departure time, or any other planned time along a target delivery route.

(68) Referring now to FIG. **13**, a process **1300** for determining spatial availability for the compartment **28** of the vehicle **12** may include scanning the vehicle **12** and the cargo **37** via the LiDAR modules **22** at step **1302**. Simultaneously, the LiDAR modules **22** may scan the region exterior **20** to the vehicle **12** to detect and map any objects **138** to be loaded into the compartment **28**. At step **1304**, dynamic free space in the compartment **28** of the vehicle **12** is defined using the at least one point cloud **24** generated by the LiDAR module **22**. At step **1306**, a morphing level for the compartment **28**, such as a dynamic morphing level, a static morphing level, or a pseudo-static morphing level is defined by the detection system **10**. The morphing level may be selected by a user or otherwise suggested or determined by the processing circuitry **40**. The morphing level is applied to the dynamic free space in the compartment **28** as mapped by the LiDAR module **22** at step **1308**. For example, requisite mass, weight, volume, or general spatial availability may be determined for feasibility to morph the compartment **28**. For example, a first level of availability may be determined based on the feasibility of volumetric changes to the free space. At step **1310**, a second level of availability may be defined based on the remaining free space after the morphing conditions are applied. For example, parameters related to minimum spacing between occupants **26**, the number of occupants **26** in the compartment **28**, health parameters pertaining to individual occupants **26**, or other distancing requirements may be manually entered, determined by the processing circuitry **40**, or otherwise incorporated into the detection system **10** to further refine the availability at the second availability level for determining the feasibility of the morphing level.

(69) Once a final availability is determined by the detection system **10**, the processing circuitry **40** may limit the options for the particular morphing conditions to a number of options **182** at step **1312**. The processing circuitry **40** may rank or order the number of options **182** based on a preferred user parameter, such as style, stability, or space. For example, different target configurations may have different levels of optimization. As previously described, and by way of example, if a passenger cabin **32** is intended to be transformed to have a kitchen/living area **168**, the particular kitchen/living arrangement selected by the user may be one of a plurality of options **182** that are available. A first option for the kitchen/living area **168** may prioritize style (e.g., color, geometries, materials employed for fixtures). A second option may optimize stability for maneuvering and controlling the vehicle **12**. In this way, the second option may prioritize load distribution normalization. For example, the second option may locate the kitchen and living area **168** features centrally along a median line of the vehicle **12** or peripherally to allow for the center of mass to be equally displaced in the compartment **28**. A third option that prioritizes space may

further be offered for selection, which may recommend smaller fixtures (e.g., smaller sinks, a smaller bed, a smaller seating arrangement). As previously described, the options **182** may also be limited or prioritized by access to electrical and/or water, or other utility accesses. For example, if heating, ventilation, and air conditioning (HVAC) connections are distributed toward the central portion of the compartment **28**, walking spaces may be available in the central areas to allow for optimal climate conditions throughout areas most commonly used in the kitchen/living area **168**. It is contemplated the kitchen/living area **168** example of the present disclosure is nonlimiting and exemplary in nature. Other target configurations and/or arrangements may be requested by the user and/or determined by the processing circuitry **40** and relate to specific functions. For example, although not described in detail herein, an accessory for handicap accessibility to the compartment **28** may be a static or pseudo-static morphing level that may be analyzed by the processing circuitry **40** to determine the availability for such a modification.

(70) At step **1314**, the process **1300** may run a morphing optimization to select the optimal arrangement for the given parameters entered by the user and/or selected by the processing circuitry **40**. Based on the optimization, the processing circuitry **40** may be configured to recommend the optimized morphing arrangement at step **1316**. It is contemplated that, following a morphing, such as after unloading or loading cargo **37**, permanently modifying the structure of the compartment **28** (e.g., a static morphing), or temporarily modifying the structure of the vehicle **12** (e.g., a pseudo-static morphing), the process **1300** may repeat steps **1302** through **1316** in order to map the new environment for further optimization of the compartment **28** for space management.

(71) In general, the present detection system **10** may provide for effective transformability using the LiDAR modules **22**. The detection system **10** analyzes, preplans, and provides optimal schema for utilizing space in vehicles **12** based on content geometry, weight, and style, and provides real-time data of the spatial changes. Accordingly, methods and techniques applied by the detection system **10** may incorporate guidance for target configurations that offer optimal load distribution, style preferences, or any other parameter selected by the user or the processing circuitry **40**. Such optimal morphing recommendations may provide for effective power consumption due to balanced loads, optimized climate control placement relative to occupant positions, or any other parameter optimized based on the target configuration.

(72) With regard to commercial applications and/or livery/delivery services, the present detection system **10** may further provide for a recommending optimized organization for the compartment **28** based on the at least one point cloud **24** of the cargo **37** or objects to be loaded. Further, using interior or exterior interior LiDAR modules **22** or reading from specification data from a specification sheet, purchase order, or other documentation of the load, load distributions may be analyzed for optimal placement recommendations. Inventory devices, such as barcode scanners or other scanning devices may capture the specification information, which may then be compiled with the at least one point cloud **24** by the detection system **10** to optimize load distribution in the compartment **28**.

(73) Other enhancements may be provided by the detection system **10**, including adjustments for target routes to delivery or loading points as previously described. For example, enhanced scheduling may be provided based on adjustments to arrival time as determined by the processing circuitry **40**. In this way, the detection system **10** may save time by adjusting time estimates due to the load distribution and determined availability in each vehicle for a fleet operation or a single-vehicle operation. Further, the cargo **37** may further be organized by the present detection system **10** based on the type of cargo **37** (e.g., contents being liquids, glass, powder, etc.) to limit contact with the objects and or limit jostling. Accordingly, the space may be optimized to enhance securement of the cargo **37**. Further, the strength or sharpness of boxes **128** or other cargo **37** may be estimated based on the at least one point cloud **24** to allow the processing circuitry **40** to determine optimal placement. In this way, dynamic updates to the cargo compartment localization and management may effectuate change to loading/unloading and distributions as a function of

time and/or location.

(74) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A method for determining spatial availability for a compartment of a vehicle, the method comprising: generating, via at least one time-of-flight sensor, a first point cloud representing the compartment of the vehicle, the first point cloud including three-dimensional positional information of the compartment; calculating, via processing circuitry in communication with the at least one time-of-flight sensor, spatial information corresponding to a target configuration for the compartment; comparing, via the processing circuitry, the spatial information to the first point cloud; determining an availability of the target configuration based on the comparison of the spatial information to the first point cloud; identifying cargo in the compartment based on the first point cloud; and estimating a load distribution of the cargo, wherein determining the availability is based further on the load distribution.
2. The method of claim 1, further comprising: determining the target configuration based on an occupancy condition of the compartment.
3. The method of claim 2, wherein the occupancy condition includes a number of occupants and a minimum spacing between the number of occupants.
4. The method of claim 1, further comprising: determining the target configuration based on dimensional data of an object to be loaded into the compartment.
5. The method of claim 4, further comprising: generating, via the at least one time-of-flight sensor, a second point cloud representing the object to be loaded into the compartment, the second point cloud including three-dimensional positional information of the object; and calculating, via the processing circuitry, the dimensional data based on the second point cloud.
6. The method of claim 5, wherein the at least one time-of-flight sensor includes a first LiDAR module in the vehicle configured to capture the first point cloud and a second LiDAR module configured to capture the second point cloud.
7. The method of claim 4, further comprising: estimating a weight of the object based on the dimensional data, wherein determining the availability is based further on the weight of the object.
8. The method of claim 1, further comprising: determining, via the processing circuitry, an unloading plan for the cargo based on the load distribution.
9. The method of claim 8, further comprising: determining an identity of an unloading source for the cargo based on the unloading plan, wherein the identity corresponds to a work capacity of the unloading source.
10. The method of claim 8, wherein the unloading plan includes route information for the vehicle, and further comprising: determining the route information based on the load distribution.
11. The method of claim 8, further comprising: determining a timing for the unloading plan based on the load distribution.
12. The method of claim 11, wherein the timing is an arrival time for the vehicle along a target delivery route.
13. The method of claim 1, wherein the vehicle is a first vehicle of a fleet of vehicles, and further comprising: generating, via the at least one time-of-flight sensor, a second point cloud of a loading region of a second vehicle of the fleet of vehicles, the second point cloud including three-dimensional positional information of the compartment; comparing the first point cloud to the second point cloud; and determining a loading plan for the fleet of vehicles based on the comparison of the first point cloud to the second point cloud.

14. A system for determining spatial availability for a compartment of a vehicle, the system comprising: at least one time-of-flight sensor configured to generate a first point cloud representing the compartment of the vehicle, the first point cloud including three-dimensional positional information of the compartment; and processing circuitry in communication with the at least one time-of-flight sensor configured to: calculate spatial information corresponding to a target configuration for the compartment; compare the spatial information to the first point cloud; and determine an availability of the target configuration based on the comparison of the spatial information to the first point cloud, wherein the at least one time-of-flight sensor includes a first LiDAR module in the vehicle configured to capture the first point cloud and a second LiDAR module configured to capture a second point cloud of an object to be loaded into the compartment, wherein the second point cloud includes three-dimensional positional information of the object.
15. The system of claim 14 wherein the processing circuitry is further configured to determine the target configuration based on an occupancy condition of the compartment.
16. The system of claim 15, wherein the occupancy condition includes a number of occupants and a minimum spacing between the number of occupants.
17. A method for determining spatial availability for a compartment of a vehicle, the method comprising: generating, via at least one LiDAR module, a first point cloud representing an interior of the vehicle, the first point cloud including three-dimensional positional information of the interior; generating, via the at least one LiDAR module, a second point cloud representing an object to be loaded into the compartment of the vehicle, the second point cloud including three-dimensional positional information of the object; calculating, via processing circuitry in communication with the at least one LiDAR module, dimensional data of the object to be loaded based on the second point cloud; determining, via the processing circuitry, a target configuration based on the dimensional data of the object to be loaded; calculating, via the processing circuitry, spatial information corresponding to the target configuration for the compartment; comparing, via the processing circuitry, the spatial information to the first point cloud; and determining an availability of the target configuration based on the comparison of the spatial information to the first point cloud.
18. The method of claim 17, wherein the at least one LiDAR module includes a first LiDAR module in the vehicle configured to capture the first point cloud and a second LiDAR module configured to capture the second point cloud.
-